Sharukesh M

B.Tech AI & Data Science Student

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Education

Amrita Vishwa Vidyapeetham — Coimbatore, India
B.Tech, Artificial Intelligence & Data Science | GPA: 8.78/10.00 — *Aug 2023 – Oct 2027*

Sri Chaitanya Techno School — Chennai, India
High School: 83.2% | Higher Secondary: 85.6% — *Apr 2021 – Apr 2023*

Experience

Telesto Energy — Singapore (Remote)
Artificial Intelligence Intern — *Dec 2025 – Mar 2026*

- Architected a multi-modal **RAG** system using **AWS Bedrock**, **FAISS** for sub-millisecond vector search, and **Neo4j** for graph-based multi-hop retrieval, ingesting heterogeneous archival records (PDFs, scans, spreadsheets, presentations) across oil well basins.
- Engineered an **OpenCV**-based document restoration pipeline that sharpened degraded scans and removed artifacts, improving downstream OCR readability on low-quality archival images.
- Enabled domain experts to complete analysis that previously took days to months in a matter of hours by unifying multi-format retrieval into a single semantic search layer.

Discovery AI LLC — Plano, TX, USA (Remote)
Artificial Intelligence Intern — *Sep 2025 – Nov 2025*

- Built a web-based **AR Rug Visualizer** using **FastAPI** and **OpenCV** with floor plane estimation and projective geometry; adopted by Rugtown (Plano, TX) as both a customer-facing web tool and an in-store sales aid.
- Integrated **Pyodide** for browser-native Python execution, cutting per-interaction latency from 5 seconds to low milliseconds by eliminating server round-trips.

InceptAI — Coimbatore, India (Remote)
Artificial Intelligence Intern — *Mar 2025 – Apr 2025*

- Designed **Microsoft Security Copilot** plugins with **RAG pipelines** (DeepSeek R1) for context-aware KQL generation, with agentic workflows built to accelerate SOC incident response.
- Work presented at the Gartner Security & Risk Management Summit; shipped to production via **CI/CD pipelines** in collaboration with senior engineers.

Projects

Hemo-Flash: Efficient Hematology Detection *[Journal Paper in Submission]* — Jan – Mar 2026

- Engineered a lightweight **YOLOv11** variant with a BiFPN neck and Ghost convolutions for real-time RBC/WBC/platelet detection, cutting parameters by **86%** vs. SOTA YOLO while retaining **98.3% mAP**.
- Optimized inference via **TensorRT** for edge deployment, achieving **187 FPS** on Jetson Nano Orin and **25 FPS** on Raspberry Pi 5 – enabling real-time automated blood cell counting on low-power hardware.

Scalable Academic Research Engine — Aug – Oct 2025

- Engineered a distributed data pipeline using **Apache Spark** (Scala) and **HDFS** to generate SciBERT embeddings across a **1.4TB** academic paper corpus.
- Deployed a **CUDA**-accelerated **FAISS** backend delivering real-time semantic similarity search with **2–3 second** query latency at scale.

Skin Lesion Classification — Jan – Apr 2025

- Designed **RCCM**, a multi-task CNN extending U-Net, achieving **85.82%** classification accuracy and an **84.92%** Dice similarity coefficient for joint segmentation and classification.
- Built a secure full-stack app (**FastAPI**, **PostgreSQL**, JWT auth) delivering diagnosis, treatment recommendations, and geolocation-based appointment booking.

Publications

Mahendran, S. et al. "Analysis of Deep and Graph Neural Networks for Enhanced DDoS Attack Detection." *[IEEE, 2026].*
Mahendran, S. et al. "An Intelligent Plug-In Playable Orchestration Model Combining Gen-AI, RAG and Agentic AI for Cybersecurity Operations." *[Springer Nature, 2026].*

Technical Skills

Languages: Python, Java, C, SQL
AI / ML: PyTorch, TensorFlow, scikit-learn, FAISS, CUDA, Hugging Face Transformers, LangChain
LLM & RAG: RAG Pipelines, Agentic AI, Prompt Engineering
Computer Vision: OpenCV, Image Segmentation
Data & Systems: Hadoop, Neo4j (Graph DB), Redis, pandas, NumPy
Web & APIs: FastAPI, REST APIs, CI/CD Pipelines
Tools & Platforms: Git, Linux, AWS, Docker, Jupyter, VS Code
Functional Skills: Problem Solving, Team Collaboration, Cross-functional Communication, Ownership